Nümerik analiz proje ödevi 2

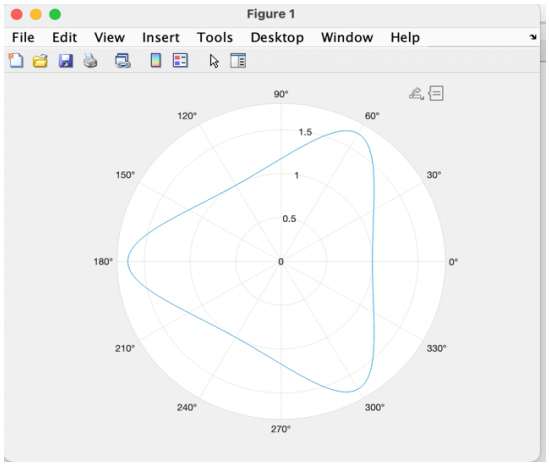
a) kodu:

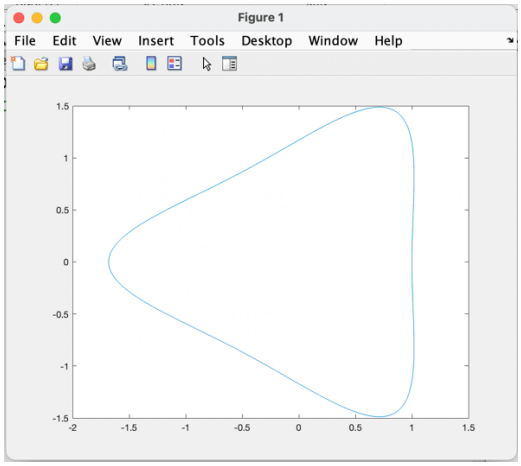
```
%%
fi = linspace(0,2*pi,360);
polarplot(fi,p(fi));
%%
fi = linspace(0,2*pi,360);
x = p(fi) .* cos(fi);
y = p(fi) .* sin(fi);
plot(x,y)

function [outputArg1] = p(fi)
%P Summary of this function goes here
%   fi is element [0,2*pi]
outputArg1 = (cos(3.*fi/4).^8 + sin(3.*fi/4).^8).^(-1/4);

end
```

Grafikler sırasıyla bu şekilde :





$$x = \rho(\emptyset)\cos(\emptyset)\ , y = \rho(\emptyset)\sin(\emptyset)$$

Birinci grafikte $\rho(\emptyset)$ fonksiyonun $\emptyset$ ' ye göre kutupsal koordinatta çizimini temsil ediyor, Kartezyen koordinatlardan kutupsal koordinatları geçmek için :

$$x = r\ sin(\emptyset) \quad y = r\cos(\emptyset)$$

$$ve\ x^2 + y^2 = r^2\ bağlantısı\ kullanılır$$

Burdan x ve y nin $\rho(\emptyset)$ fonksiyonun $\emptyset$ kutupsal koordinattaki grafiğini bulmak için bir dönüşüm olduğu gözükmektedir.

b)

kodu :

```
fi = linspace(0,2*pi,360);
x =@(fi) p(fi) .*cos(fi);
y =@(fi) p(fi) .*sin(fi);
%%
dxfi_dfi_f = forward_difference_deritave(x,fi,fi(2));
dyfi_dfi_f = forward_difference_deritave(y,fi,fi(2));
polarplot(lx(@L,dxfi_dfi_f, dyfi_dfi_f), ly(@L, dxfi_dfi_f,dyfi_dfi_f) )
%%
g_f = forward_difference_deritave(@p,fi,fi(2));
polarplot(fi, g_f)
```

```
%%
dxfi_dfi_b = back_difference_deritave(x,fi,fi(2));
dyfi_dfi_b = back_difference_deritave(y,fi,fi(2));
polarplot(lx(@L,dxfi_dfi_b, dyfi_dfi_b), ly(@L, dxfi_dfi_b,dyfi_dfi_b) )
%%
g_b = back_difference_deritave(@p,fi,fi(2));
polarplot(fi, g_b)
%%
dxfi_dfi_e = three_point_end_point(x,fi,fi(2));
dyfi_dfi_e = three_point_end_point(y,fi,fi(2));
polarplot(lx(@L,dxfi_dfi_e, dyfi_dfi_e), ly(@L, dxfi_dfi_e,dyfi_dfi_e) )
%%
g_e = three_point_end_point(@p,fi,fi(2));
polarplot(fi, g_e)
%%
dxfi_dfi_m = three_point_middle_point(x,fi,fi(2));
dyfi_dfi_m = three_point_middle_point(y,fi,fi(2));
plot(dxfi_dfi_m,dyfi_dfi_m )
%%
g_m = three_point_middle_point(@p,fi,fi(2));
polarplot(fi, g_m)

function [outputArg1] = forward_difference_deritave(f,x0, h)
%FORWARD_DİFFERENCE_DERİTAVE Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = (f(x0 + h) - f(x0))./h;
end

function [outputArg1] = back_difference_deritave(f,x0, h)
%BACK_DİFFERENCE_DERİTAVE Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = (f(x0 - h) - f(x0))/-h;
end

function [outputArg1] = three_point_end_point(f,x0, h)
%THREE_POİNT_END_POİNT Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = ((1/(2*h))) * (-3.*f(x0) + 4.*f(x0+ h) -f(x0+2.*h));
end

function [outputArg1] = three_point_middle_point(f,x0,h)
%THREE_POİNT_MİDDLE_POİNT Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = (1/(2* h)) .* (f(x0+h) -f(x0-h) );
end

function [outputArg1] = lx(L,difxfi,difyfi)
%UNTİTLED Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = (1./L(difxfi,difyfi)).*difxfi;
end

function [outputArg1] = ly(L,difxfi,difyfi)
%UNTİTLED Summary of this function goes here
%   Detailed explanation goes here
outputArg1 = (1./L(difxfi,difyfi)).*difyfi;
end

function [outputArg1] = L(difxfi,difyfi)
%L Summary of this function goes here
```

```
%   Detailed explanation goes here
outputArg1 = sqrt(difxfi.^2 + difyfi.^2);
end
```

İleri fark formülü bu şekildedir:

$$f'(x) = \frac{f(x0+h) - f(x0)}{h}$$

Hatası ise bu şekilde değerlendirebilir:

$$Hata \leq |\frac{h \times f''(\zeta)}{2}|$$

Geri fark formülü:

$$f'(x) = \frac{f(x0-h) - f(x0)}{-h}$$

Hatası ise ileri fark ile aynıdır:

$$Hata \leq |\frac{h \times f''(\zeta)}{2}|$$

3 nokta uç nokta formülü :

$$f'(x0) = \frac{1}{2h}[-3f(x0) + 4f(x1) - f(x2)]$$

$$Hata \leq |\frac{f(\zeta) \times 2h^2}{6}|$$

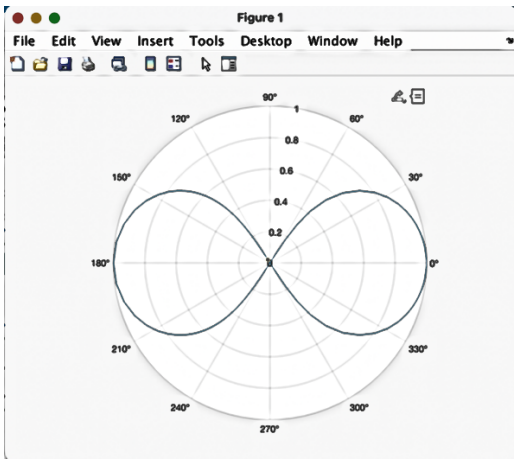
3 nokta orta nokta formülü :

$$f'(x0) = \frac{1}{2h}[f(x2) - f(x0)]$$

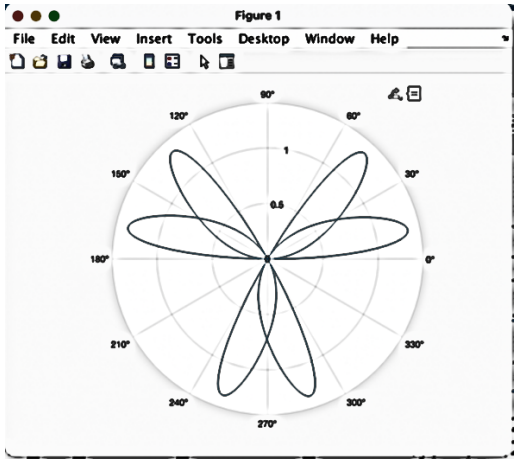
$$Hata \leq |\frac{f(\zeta) \times "h^2}{6}|$$

1) ileri fark:

ly nin lx ‘e göre grafiği

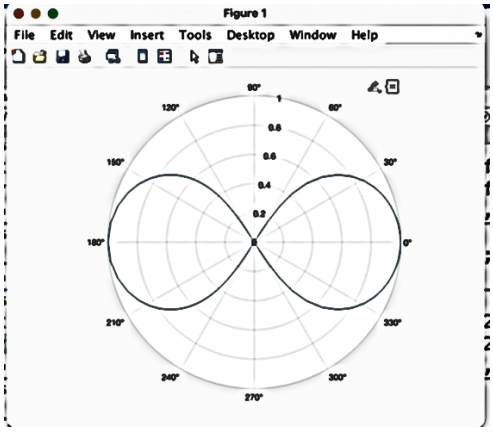


g nin fi ye göre grafiği

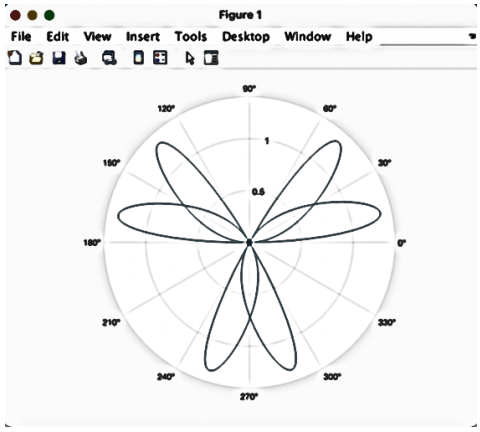


2) geri fark:

ly nin lx ‘e göre grafiği

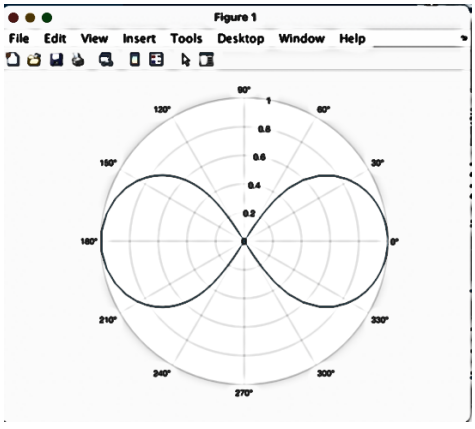


g nin fi ye göre grafiği

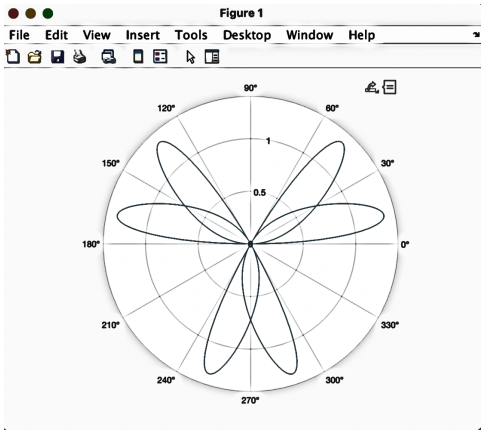


3) üç nokta son nokta

ly nin lx ‘e göre grafiği

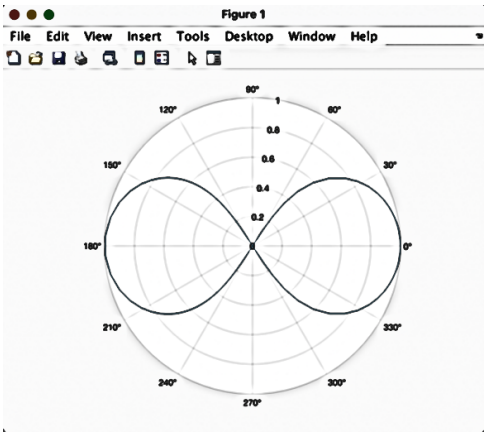


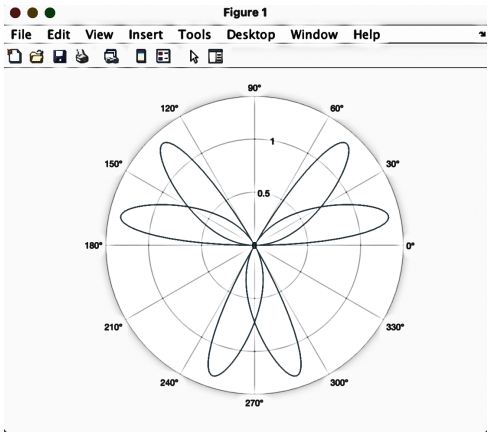
g nin fi ye göre grafiği



4) üç nokta orta nokta
ly nin lx ‘e göre grafiği

g nin fi ye göre grafiği





c) ρ fonksiyonunu $\emptyset$ ye göre çizdirdim ve ρ fonksiyonun periyodu yaklaşık 2.08 olacak şekilde periyodik bir fonksiyon olduğunu gözlemledim.

Kod:

```
fi = linspace(0,2*pi,360);
plot(fi,p(fi));
%%
syms a0 a1 a2 a3 ;
H3 = @(x) a0 + a1.*x + a2.*x.^2 + a3*x.^3;
H3_d =@(x) a1 + a2.*2.*x + a3.*3.*x.^2 ;
N = 2;
fi = linspace(0,2.08,N+1);

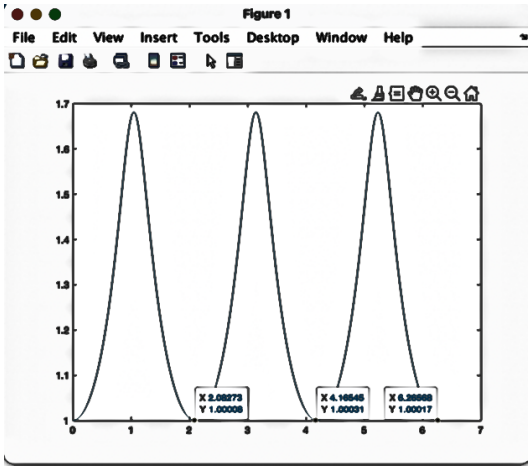
H ;
for i =1 :N
    a = fi(i);
    b = fi(i+1);
    hs = solve([H3(a)==p(a),
H3_d(a)==three_point_middle_point(@p,a,0.000000001),...
        H3(b) ==p(b) , H3_d(b) ==
three_point_middle_point(@p,b,0.000000001)]...
        ,[a0, a1, a2, a3]);

    H{1,i} = double([hs.a0 , hs.a1 ,hs.a2 ,hs.a3]);
end
fis = linspace(0,2*pi,360);
y= [];
for i=1:length(fis)
    if fis(i) < 1.04 &&fis(i) >=0
        y(i) = sum(cell2mat(H(1,1)) .* [1,fis(i),fis(i)^2,fis(i)^3],"all");
    elseif fis(i) < 2.08 &&fis(i) >1.04
        y(i) = sum(cell2mat(H(1,2)) .* [1,fis(i),fis(i)^2,fis(i)^3],"all");
    elseif fis(i) < 3.12 &&fis(i) >2.08
        y(i) = sum(cell2mat(H(1,1)) .* [1,fis(i)-2.08,(fis(i)-
2.08)^2,(fis(i)-2.08)^3],"all") ;
    elseif fis(i) < 4.16 &&fis(i) >3.12
        y(i) = sum(cell2mat(H(1,2)) .* [1,fis(i)-2.08,(fis(i)-
2.08)^2,(fis(i)-2.08)^3],"all") ;

    elseif fis(i) < 5.20 &&fis(i) >4.16
        y(i) = sum(cell2mat(H(1,1)) .* [1,fis(i)-2*2.08,(fis(i)-
2*2.08)^2,(fis(i)-2*2.08)^3],"all") ;
    elseif fis(i) < 6.28 &&fis(i) >5.20
        y(i) = sum(cell2mat(H(1,2)) .* [1,fis(i)-2*2.08,(fis(i)-
2*2.08)^2,(fis(i)-2*2.08)^3],"all") ;
```

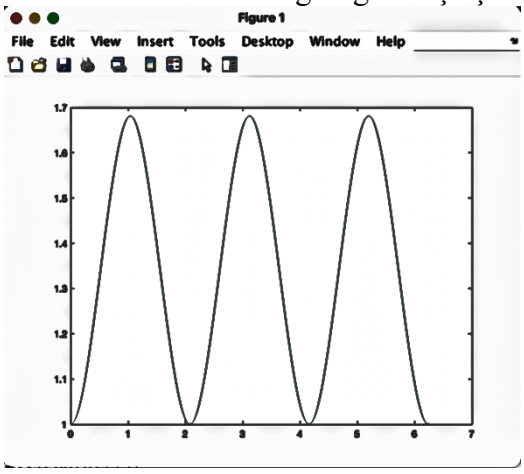
```
    end

end
plot(fis(1:359),y)
%% hata
plot(fis(1:359), (y - p(fis(1:359) )) ./ p(fis(1:359)))
```



Bu yüzden bir periyodunu aldım ve iki aralığa böldüm (çukurdan tepeye , tepeden çukura olacak şekilde) sonrasında iki tane hermit polinomu oturtturdum. Bu hermit polinomları bana bir periyodun grafiklerini verdi. Onun için diğer Ø değerleri için periyod kaydırark fonksiyonun devamını elde ettim, burda 3 kez tekrarlandığı için 3 periyodunu hesapladım.

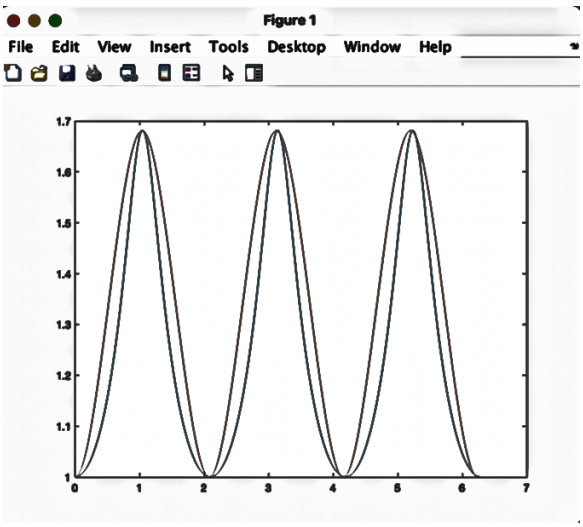
Hermit sonucu elde ettiğim grafik şu şekildedir :

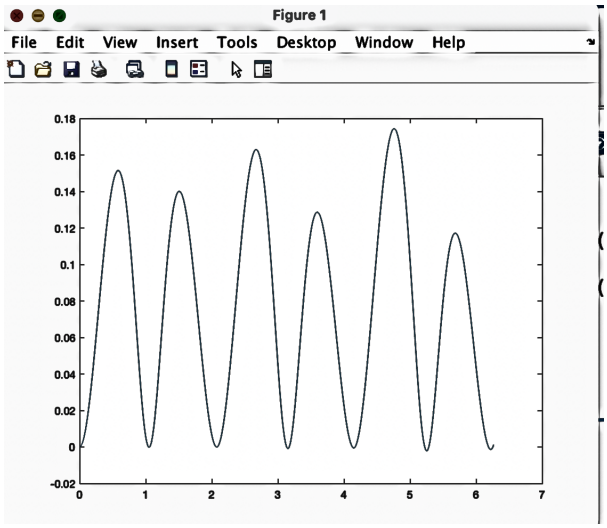


Grafiklerin benzerliğine rağmen Hermit metodu ile otutturduğumuz fonksiyonunun asıl fonksiyonumuzla tam uyuşmadığu gözükmektedir. Çukur tepe arası y nin değişimi asıl fonksiyonumuzda ilk başta yavaş sonra daha ani gözükmektedir. Kıvrım şeklinden hatayı anlayabiliriz.

İki fonksiyonu üst üste çizdirirsek aşağıdaki gibi bir şekil elde ederiz burda altta kalan fonksiyon asıl fonksiyonumuz, üstteki ise Hermit ile bulduğumuz fonksiyondur.

Hatasını ise aşağıdaki gibidir:





Yukarıdaki durum özel periyodiklikten dolayı özel bir durum sayılabilir. Bunun göz önüne bulundurmadan N(aralık sayısı) in değişimine göre oturan polinomların davranışlarına bakalım.

Kod :

```
fi = linspace(0,2*pi,360);
plot(fi,p(fi));
hold on;
%%
N = 20;
interval = [0 2*pi];
fis = linspace(0,2*pi,360);
y = Hermit(@p, interval ,fis,N );

plot(fis,y)
%% hata
%plot(fis(1:359), (y - p(fis(1:359) )) ./ p(fis(1:359)))

function [outputArg1] = Hermit(p,interval,fi,N )
%HERMİT Summary of this function goes here
%   N is the number of intervals
interval_points= linspace(interval(1), interval(2), N);
syms a0 a1 a2 a3 ;
H3 = @(x) a0 + a1.*x + a2.*x.^2 + a3*x.^3;
H3_d =@(x) a1 + a2.*2.*x + a3.*3.*x.^2 ;

for i =1 :N -1
    a = interval_points(i);
    b = interval_points(i+1);
    hs = solve([H3(a)==p(a),
H3_d(a)==three_point_middle_point(@p,a,0.000000001),...
        H3(b) ==p(b) , H3_d(b) ==
three_point_middle_point(@p,b,0.000000001)]...
        ,[a0, a1, a2, a3]);

    H{1,i} = double([hs.a0 , hs.a1 ,hs.a2 ,hs.a3]);
end

y= [];
for i=1:length(fi)
    for j= 1: length(interval_points) -1
        if fi(i) <= interval_points(j+1) && fi(i) >=interval_points(j)
```

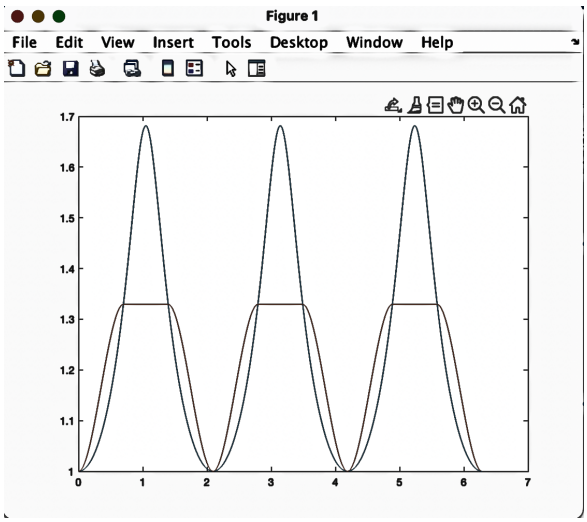
```
            y(i) = sum(cell2mat(H(1,j)) .*
[1,fi(i),fi(i)^2,fi(i)^3],"all");
        end
    end

outputArg1 = y;

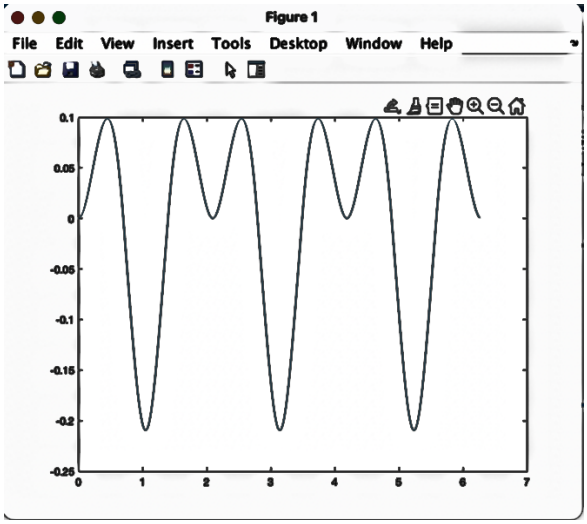
end
```

N = 10 için

Fonksiyonlar

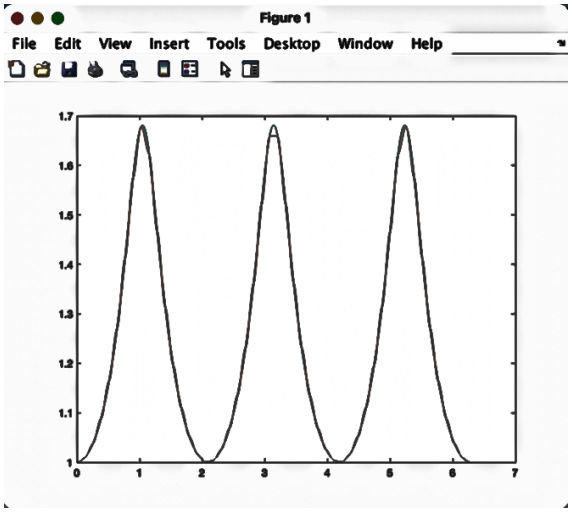


Hata

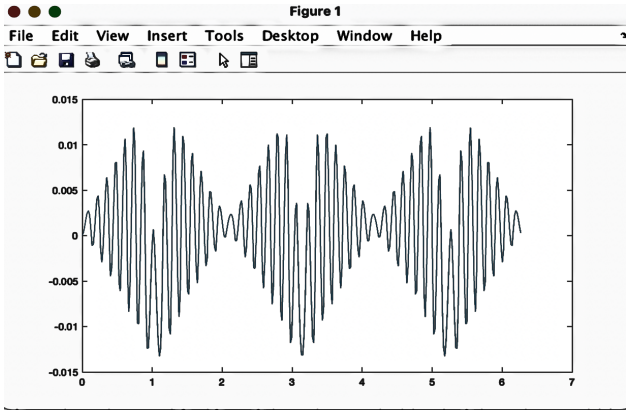


N = 50 için

Fonksiyonlar



Hata

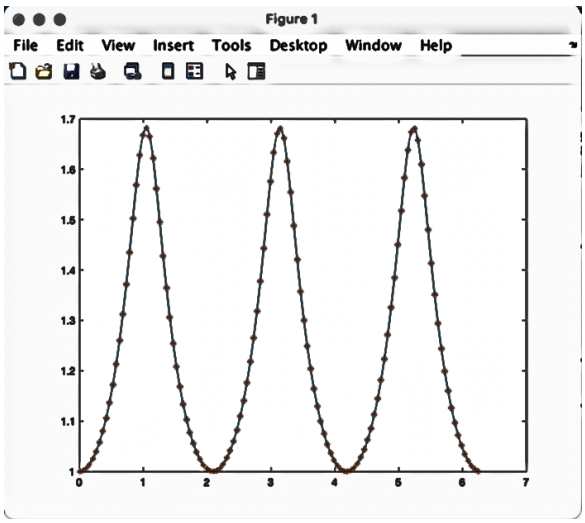


N = 360 için

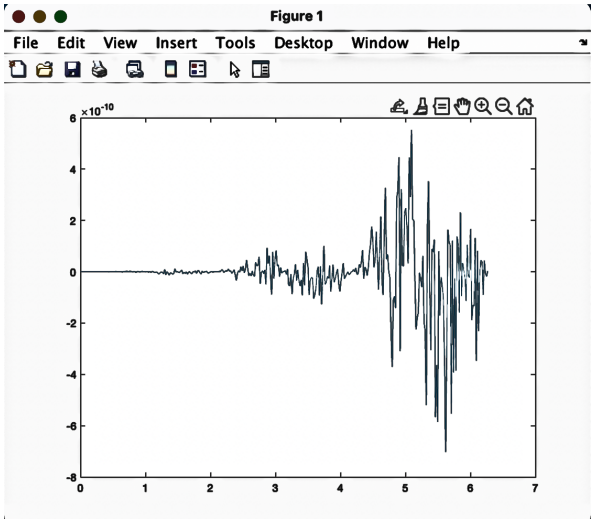
Kod :

```
fi = linspace(0,2*pi,360);
plot(fi,p(fi));
hold on;
%%
N = 360;
interval = [0 2*pi];
fis = linspace(0,2*pi,360);
y = Hermit(@p, interval ,fis,N );
%%
plot(fis,y,'*','MarkerIndices',1:3:length(y))
```

Fonksiyonlar : (hermit yıldızlı)



Hata



Hermit fonksiyonun hatasını değerlendirmek için bu denklem kullanılır

H_{2n+1} *için*

$$HATA \leq \left| \frac{f^{2n+2}(\zeta)}{(2n+2)!} (x-x_0) \times (x-x_1) \dots (x-x_n) \right|$$

Burda H 3üncü dereceden o yüzden ve f = rho o yüzden hata böyle olur:

$$HATA \leq \left| \frac{\rho^4(\zeta)}{24} (x-x_0) \times (x-x_1) \times (x-x_2) \right|$$

d)kodu :

```
fi = linspace(0,2*pi,360);
y_h = Hermit(@p,[0,2*pi],fi,360);
xh =@(fi) y_h .*cos(fi);
yh =@(fi) y_h .*sin(fi);
%%
p_fi = p(fi);
x =@(fi) p_fi .*cos(fi);
y =@(fi) p_fi .*sin(fi);
%%
t = tiledlayout(2,4);
dxfi_dfi_f_h = forward_difference_deritave(xh,fi,fi(2));
dyfi_dfi_f_h = forward_difference_deritave(yh,fi,fi(2));
dxfi_dfi_f = forward_difference_deritave(x,fi,fi(2));
```

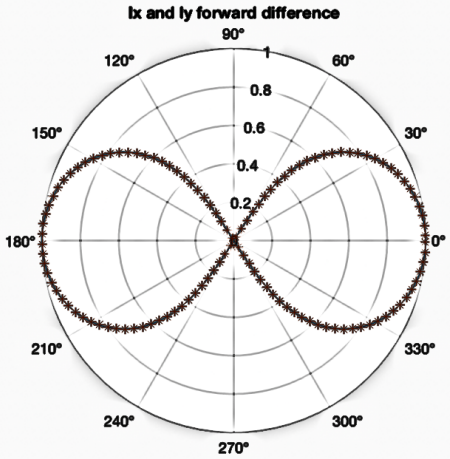
```
dyfi_dfi_f = forward_difference_deritave(y,fi,fi(2));
nexttile
%%
polarplot(lx(@L,dxfi_dfi_f_h, dyfi_dfi_f_h), ly(@L,
dxfi_dfi_f_h,dyfi_dfi_f_h) )
hold on
polarplot(lx(@L,dxfi_dfi_f, dyfi_dfi_f), ly(@L, dxfi_dfi_f,dyfi_dfi_f)
,"*",'MarkerIndices',1:3:length(ly(@L, dxfi_dfi_f,dyfi_dfi_f)))
%%
%polarplot(lx(@L,dxfi_dfi_f, dyfi_dfi_f)  , ly(@L,
dxfi_dfi_f_h,dyfi_dfi_f_h) -ly(@L, dxfi_dfi_f,dyfi_dfi_f) ./ ly(@L,
dxfi_dfi_f,dyfi_dfi_f) )
title("lx and ly forward difference")
%%
h= fi(2);
g_f_h = ( y_h(2:360) - y_h(1:359)  )./h;
g_f = forward_difference_deritave(@p,fi,fi(2));
nexttile
polarplot(fi(1:359), g_f_h)
hold on
polarplot(fi , g_f,"*",'MarkerIndices',1:2:length(ly(@L,
dxfi_dfi_f,dyfi_dfi_f)) )
title("g forward difference")
%%
%%%%%%%%%%%%%%%%
dxfi_dfi_b_h = back_difference_deritave(xh,fi,fi(2));
dyfi_dfi_b_h = back_difference_deritave(yh,fi,fi(2));

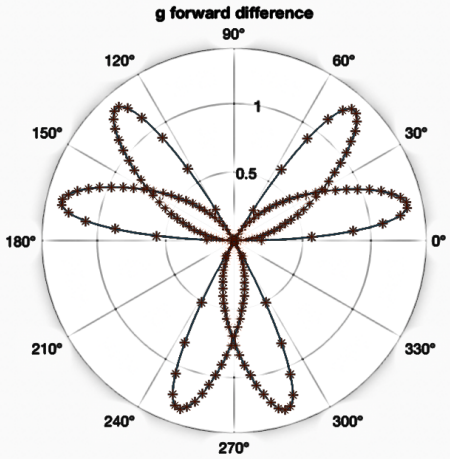
dxfi_dfi_b = back_difference_deritave(x,fi,fi(2));
dyfi_dfi_b = back_difference_deritave(y,fi,fi(2));
nexttile
polarplot(lx(@L,dxfi_dfi_b_h, dyfi_dfi_b_h), ly(@L,
dxfi_dfi_b_h,dyfi_dfi_b_h) )
hold on
polarplot(lx(@L,dxfi_dfi_b, dyfi_dfi_b), ly(@L, dxfi_dfi_b,dyfi_dfi_b)
,"*",'MarkerIndices',1:5:length(ly(@L, dxfi_dfi_f,dyfi_dfi_f)))

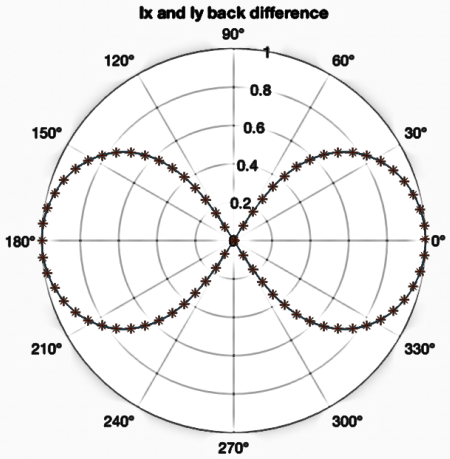
title("lx and ly back difference")
%%
%%%%%%%%%%%%%%%
h= fi(2);
g_b_h = (  y_h(1:359) - y_h(2:360)  )./ -h;
g_b = back_difference_deritave(@p,fi,fi(2));
nexttile
polarplot(fi(2:360), g_b_h)
hold on
polarplot(fi, g_b,"*",'MarkerIndices', 1:2:360 )
title("g back difference")
%%
%%%%%%%%%%%%%%%
dxfi_dfi_e_h = three_point_end_point(xh,fi,fi(2));
dyfi_dfi_e_h = three_point_end_point(yh,fi,fi(2));
dxfi_dfi_e = three_point_end_point(x,fi,fi(2));
dyfi_dfi_e = three_point_end_point(y,fi,fi(2));
nexttile
polarplot(lx(@L,dxfi_dfi_e_h, dyfi_dfi_e_h), ly(@L,
dxfi_dfi_e_h,dyfi_dfi_e_h) )
hold on
polarplot(lx(@L,dxfi_dfi_e, dyfi_dfi_e), ly(@L, dxfi_dfi_e, dyfi_dfi_e),
"*",'MarkerIndices', 1:4:360)
```

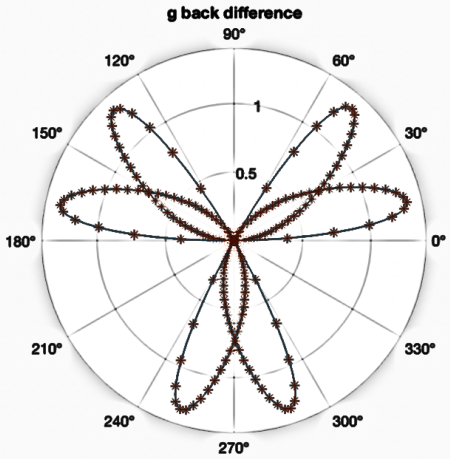
```
title("lx and ly three point end point")
%%
%%%%%%%%%%%%%
h= fi(2);
g_e_h = (1/(2*h)) * (-3.*y_h(1:358) + 4.*y_h(2:359) -y_h(3:360));
g_e = three_point_end_point(@p,fi,fi(2));
nexttile
polarplot(fi(1:358), g_e_h)
hold on
polarplot(fi, g_e, "*",'MarkerIndices', 1:2:360 )
title("g three point end point")
%%
%%%%%%%%%%%%
dxfi_dfi_m_h = three_point_middle_point(xh,fi,fi(2));
dyfi_dfi_m_h = three_point_middle_point(yh,fi,fi(2));
dxfi_dfi_m = three_point_middle_point(x,fi,fi(2));
dyfi_dfi_m = three_point_middle_point(y,fi,fi(2));
nexttile
%%
polarplot(lx(@L,dxfi_dfi_m_h, dyfi_dfi_m_h), ly(@L,
dxfi_dfi_m_h,dyfi_dfi_m_h) )
hold on
polarplot(lx(@L,dxfi_dfi_m, dyfi_dfi_m), ly(@L, dxfi_dfi_m, dyfi_dfi_m)
,"*","MarkerIndices",1:4:360 )
title("lx and ly middle point")
%%
%%%%%%%%%%%
h= fi(2);
g_m_h = (1/(2* h)) * (y_h(3:360) -y_h(1:358) );
g_m = three_point_middle_point(@p,fi,fi(2));
nexttile
polarplot(fi(2:359), g_m_h)
hold on
polarplot(fi, g_m,"*","MarkerIndices",1:4:360)
title("g middle point")
```

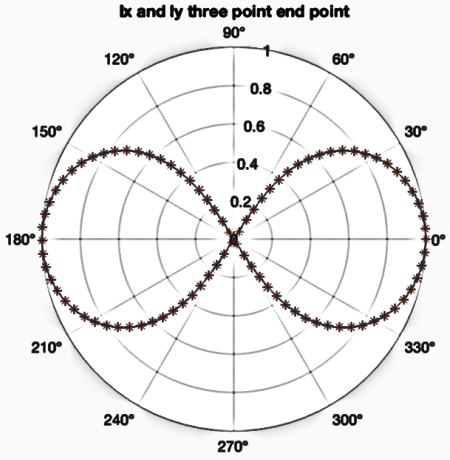
Aşağıdaki fonksiyonlar çizdirilmiştir yıldızlı olanlar Hermitli fonksiyonları temsil etmektedir.

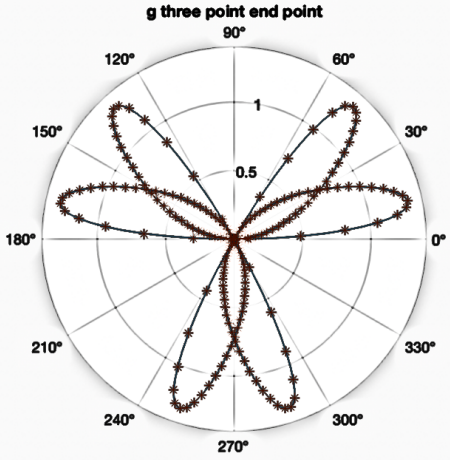

lx and ly forward difference
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.8
0.6
0.4
0.2

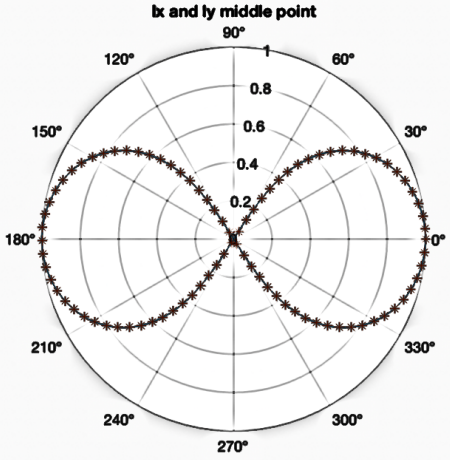

g forward difference
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.5

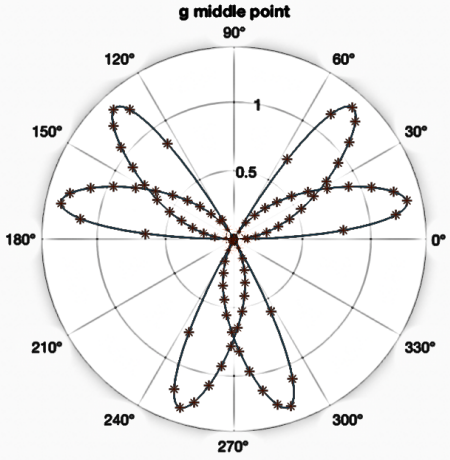

lx and ly back difference
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.8
0.6
0.4
0.2


g back difference
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.5

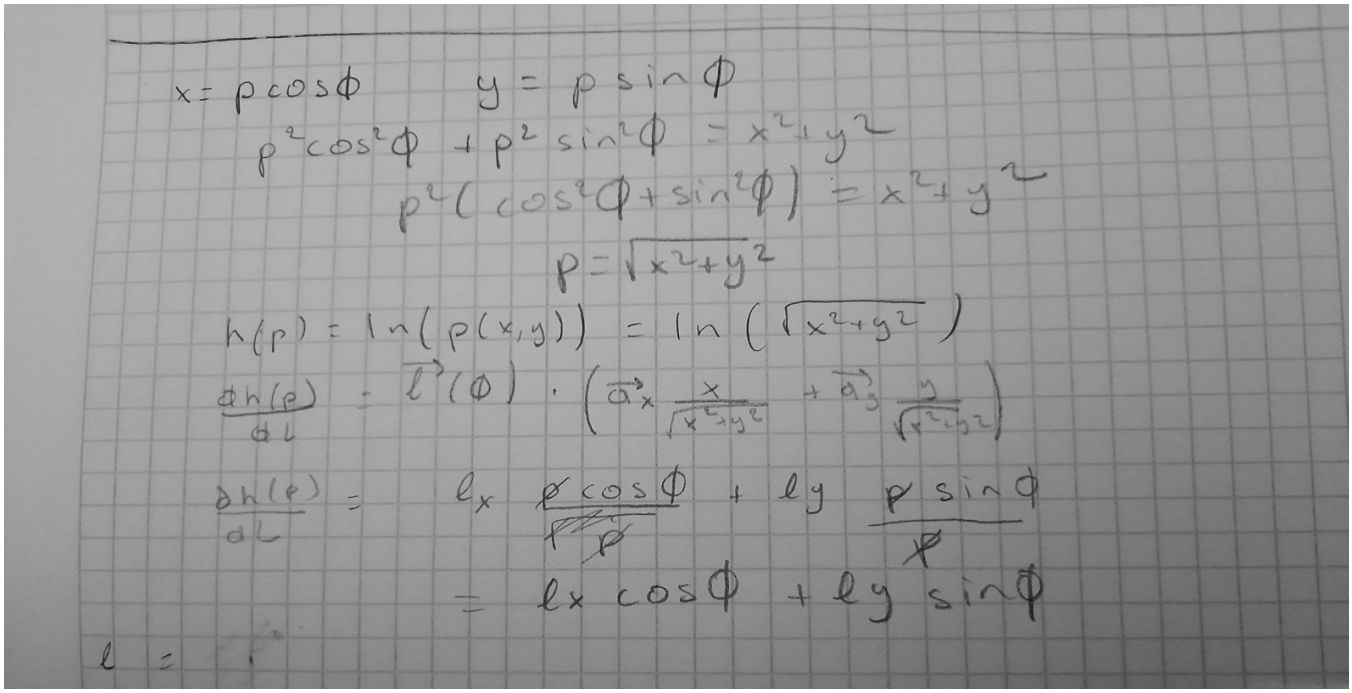

lx and ly three point end point
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.8
0.6
0.4
0.2


g three point end point
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.5


lx and ly middle point
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.8
0.6
0.4
0.2


g middle point
90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
1
0.5

e) Analitik işlemler :

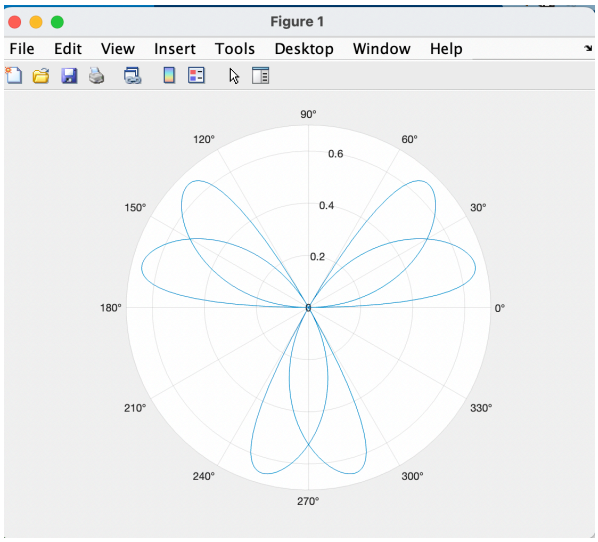


Kodu:

ρ için:

```
fi = linspace(0,2*pi,360);
x_p =@(fi) p(fi) .*cos(fi);
y_p =@(fi) p(fi) .*sin(fi);
dxfi_dfi_m_p = three_point_middle_point(x_p,fi,fi(2));
dyfi_dfi_m_p = three_point_middle_point(y_p,fi,fi(2));
dh =@(L,fi,difxfi, difyfi) lx(L,difxfi, difyfi) .* cos(fi)...
    + ly(L,difxfi, difyfi) .* sin(fi);
valueofdh_p=dh(@L,fi,dxfi_dfi_m,dyfi_dfi_m);
polarplot(fi,valueofdh_p)
```

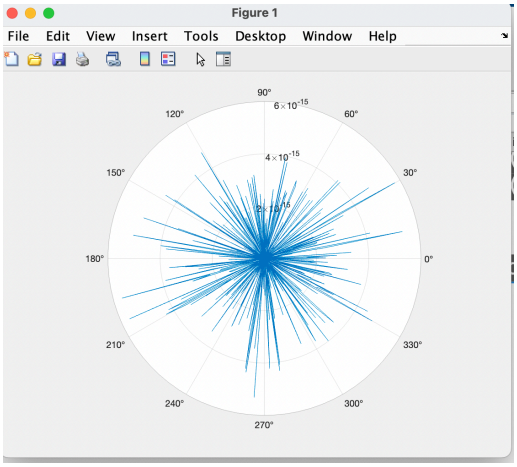
Bu analitik işlemler sonucu h fonksiyonun türevi tanımlanmış ve sırasıyla p fonksiyonu ve Hermit için çizdirilmiştir.



Hermit polinomunu çizdirirken ise böyle garip bir harekete rastlandı

Kod :

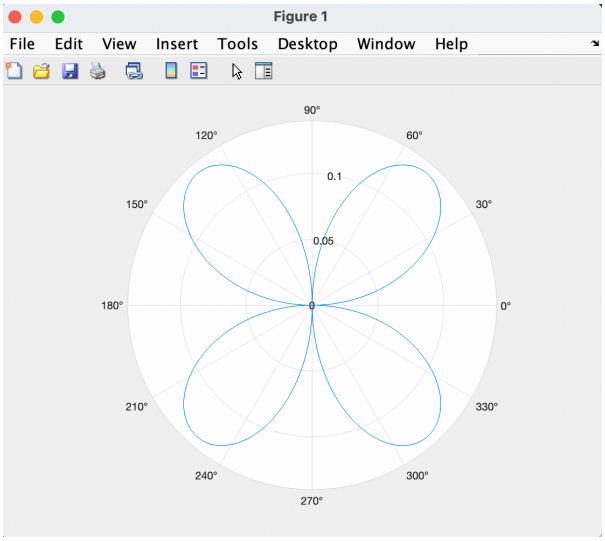
```
fi = linspace(0,2*pi,360);
dh =@(L,fi,difxfi, difyfi) lx(L,difxfi, difyfi) .* cos(fi)...
    + ly(L,difxfi, difyfi) .* sin(fi);
y_h_n = Hermit(@p,[0,2*pi],fi,360);
xh =@(fi) y_h_n .* cos(fi);
yh =@(fi) y_h_n .* sin(fi);
%%
dxfi_dfi_m_h_n = three_point_middle_point(xh,fi,fi(2));
dyfi_dfi_m_h_n = three_point_middle_point(yh,fi,fi(2));
valueofdh_h=dh(@L,fi,dxfi_dfi_m_h_n,dyfi_dfi_m_h_n);
%%
polarplot(fi,valueofdh_h)
```



Türev hassasiyetini değiştirince ise başka grafik elde ettim

Kod :

```
fi = linspace(0,2*pi,360);
dh =@(L,fi,difxfi, difyfi) lx(L,difxfi, difyfi) .* cos(fi)...
    + ly(L,difxfi, difyfi) .* sin(fi);
y_h_n = Hermit(@p,[0,2*pi],fi,360);
%%
xh =@(fi) y_h_n .* cos(fi);
yh =@(fi) y_h_n .* sin(fi);
%%
dxfi_dfi_m_h_n = three_point_middle_point(xh,fi,fi(2));
dyfi_dfi_m_h_n = three_point_middle_point(yh,fi,0.02);
valueofdh_h=dh(@L,fi,dxfi_dfi_m_h_n,dyfi_dfi_m_h_n);
%%
polarplot(fi,valueofdh_h)
```



Bunun neyden kaynaklandığını tam anlaşılamadı hassasiyetin böyle bir hataya neden açmış olabileceğini düşünülmektedir.

Bu projenin sonucunda ileri fark, ger fark, üç nokta ara nokta, ve üç nokta son nokta yöntemleri ile nümerik türev almayı ve örneklenmiş veriler üzerinde hermit polinomları oturtturup denklemlere dökmeyi öğrendim.